DialectGuard: Evaluating Response-Risk-Triggered Retrieval for Factual Question Answering Across Arabic Varieties

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Abstract

Arabic factual questions can be expressed in Modern Standard Arabic and multiple regional dialects, yet response-risk-triggered retrieval may introduce substantial overhead without reliably improving correctness. This paper presents DialectGuard, a heuristic framework that assesses cross-model disagreement and non-answer behavior before selectively normalizing, retrieving evidence, and regenerating answers. The evaluation comprises 65 development questions and 100 held-out base questions, each rendered in four Arabic varieties. Across 4,020 recorded experimental outputs, 1,600 correspond to the four-condition held-out comparison. DialectGuard intervened on 378 of 400 variants (94.5%). Its recorded median latency was 16.49 s, compared with 7.70 s for always-on retrieval, and it was slower on 399 of 400 paired examples. Provisional, AI-assisted factual labels showed no observed paired correctness advantage over always-on retrieval on 318 adjudicable pairs (difference 0.0 percentage points; clustered 95% interval −5.8 to +6.0). Unresolved factual and evidence-support judgments preclude a definitive claim of factual superiority or hallucination reduction. The results identify the cost and selectivity challenges of multi-model risk assessment in Arabic factual question answering.

Keywords: Arabic dialects; factual question answering; adaptive retrieval; retrieval-augmented generation; uncertainty; evaluation

1 Introduction

Large language models (LLMs) can generate fluent responses to factual queries while still making errors, abstaining, or introducing unsupported details. Retrieval-augmented generation (RAG) aims to provide a source of contextual evidence, but retrieving passages and regenerating responses imposes computational and engineering costs. Selective retrieval is attractive only if it reliably detects cases that need intervention and avoids paying for unnecessary processing on low-risk inputs.

Arabic introduces an additional challenge. A factual intent may be expressed in Modern Standard Arabic (MSA), Jordanian Arabic, Egyptian Arabic, or Saudi Arabic, and model behavior need not be invariant to that variation. Lexical choices, transliteration, morphology, and dialect-specific phrasing also complicate automatic exact-match evaluation. An answer such as جوبيتر can be semantically equivalent to the reference answer المشتري, while a longer answer can contain both a correct entity and an unrelated false assertion. Thus, practical analysis must distinguish answer-level correctness from claim-level evidential support.

Prior work has evaluated Arabic hallucination phenomena and dialect-related benchmark performance (Alansari & Luqman, 2025; Altakrori et al., 2026). Adaptive-RAG selects different retrieval strategies by question complexity (Jeong et al., 2024), whereas Self-RAG and FLARE explore retrieval and critique during generation (Asai et al., 2024; Jiang et al., 2023). DialectGuard instead tests whether inexpensive-to-compute response-risk indicators are selective enough to justify an intervention pathway. Importantly, the implementation studied here uses multiple language-model responses for risk assessment; whether this is actually inexpensive is an empirical question, not a design assumption.

The study addresses five questions: (RQ1) how observed behavior varies across four Arabic varieties; (RQ2) whether disagreement and non-answer signals identify baseline errors; (RQ3) whether MSA normalization improves responses; (RQ4) how retrieved generation compares to generation without retrieval; and (RQ5) whether risk-triggered mitigation improves the quality–latency trade-off. The contributions are an auditable multi-condition experimental implementation, a paired held-out evaluation with explicit missing-judgment bounds, and an empirical account of a negative efficiency result. The study does not assert an independently verified factuality improvement.

2 Related Work

2.1 Arabic dialect evaluation

Arabic language variation has implications for the construction of equivalent prompts and their

evaluation. DialectalArabicMMLU focuses attention on dialect-sensitive language-model testing (Altakrori et al., 2026). Although the present dataset was organized by base question to preserve comparable factual intent, independent expert verification of every dialect adaptation and complete certification of benchmark independence were not documented. These limitations constrain claims about broad dialect generalization.

2.2 Factuality and Arabic hallucinations

AraHalluEval proposed a fine-grained framework for examining hallucinations in Arabic generation (Alansari & Luqman, 2025). Its emphasis on evaluation granularity is relevant because one-word answer correctness is not equivalent to full-response factual faithfulness. In this work, a subset of answers was triaged using AI assistance and rule-based alias identification, while claim-level evidence support remained incompletely adjudicated. Accordingly, the reported factual results are conditional on available labels and must not be mistaken for independently verified hallucination rates.

2.3 Retrieval scheduling and reflection

Adaptive-RAG chooses a retrieval strategy according to predicted question complexity (Jeong et al., 2024); Self-RAG conditions generation on retrieval and critique (Asai et al., 2024); and FLARE retrieves information proactively during generation (Jiang et al., 2023). DialectGuard differs principally in its routing signal: observed response disagreement and non-answer behavior. Yet a response-driven router can require multiple expensive forward passes, offsetting the savings expected from skipping retrieval. The present study tests precisely that trade-off.

3 Methods

3.1 Benchmark design

The development partition comprised 65 base factual questions, each represented in MSA, Jordanian, Egyptian, and Saudi Arabic (260 variants). The held-out partition comprised 100 base questions in the same four varieties (400 variants), yielding 1,600 final responses under four principal conditions. The held-out prompts were designated review-locked v0.1 and not used for the reported development tuning. The four variants of a base question were treated as dependent observations for uncertainty analysis. The question inventory originated from a source-derived Arabic factual QA construction process, with manual filtering and candidate-review steps; however, complete independent dialect validation and full semantic-overlap adjudication were not certified.

3.2 Models and retrieval

The target generator was Qwen3-4B-4bit. Gemma-3-4B-it-4bit and Granite-4.0-H-3B-ar provided comparison answers for the routing heuristic, and Granite generated evidence-grounded answers in retrieved conditions. Multilingual sentence embeddings and a cross-encoder reranker supported passage selection. The development-frozen evidence collection contained 37 relevance-screened Arabic Wikipedia passages from 31 source URLs. Relevance screening was not an independent factual audit, and the limited corpus did not guarantee appropriate evidence coverage of the held-out set. Model revision identifiers and execution configurations are preserved in the repository.

3.3 Intervention pipeline and conditions

Four conditions were compared: (i) baseline: Qwen answered the original prompt without retrieval; (ii) normalization: the prompt was converted to MSA prior to Qwen answering; (iii) always-RAG: retrieval and reranking preceded Granite-based grounded generation; and (iv) DialectGuard: model-disagreement and Qwen non-answer indicators triggered MSA normalization, retrieval, and Granite regeneration for high-risk prompts, while low-risk prompts retained the baseline answer. The routing heuristic was thresholded at a recorded value of one; the score was not calibrated as a factual error probability.

The primary baseline–RAG comparison changed both retrieval use and the answer-generator identity (Qwen versus Granite). To help assess this confound, separate controlled-Qwen experiments were recorded on development prompts. These are supporting experiments, not a substitute for a held-out same-generator factorial comparison.

Table 1: Completed experiment groups.

Experiment	Rows
Development, four conditions	1,040
Controlled Qwen RAG	520
Ablations	780
Efficiency benchmark	80
Held-out, four conditions	1,600
Total	4,020

3.4 Execution and integrity

The experiments ran locally on an Apple Silicon M2 machine with 16 GB unified memory, using sequential model loading to reduce memory

pressure. The held-out inference used frozen commit 12cd956c5a1afc11b3a156c1e64f8d863e97a09a and recorded output manifests. The held-out table had 400 entries for each primary condition, no duplicated question-condition keys, and no reported execution errors. Eight Granite-generated final answers were empty (five always-RAG and three DialectGuard), despite successful run status; these were retained as non-correct outputs. Exact local runtime measurements were affected by loading and caching choices, and did not include all setup costs.

3.5 Correctness labels and evidential support

Responses underwent AI-assisted semantic triage, subsequent rule-based question-specific reassignment, and review of an earlier annotation iteration by a researcher. Final proposed classifications were correct, incorrect, abstention, empty, or ambiguous. Later rule-based changes were not independently certified. Among the 1,600 responses, 278 remained ambiguous in the latest proposed dataset. No complete independent claim-level verification had been finished, so answer correctness must be distinguished from hallucination or evidence-support judgments.

For condition c, adjudicable accuracy was computed as correct responses divided by 400 minus ambiguous responses, counting abstentions and empties as not correct. To quantify the effect of unknown labels, lower and upper full-sample accuracy bounds counted ambiguous answers as all incorrect and all correct, respectively. For paired condition contrasts, only question variants with determinate labels in both conditions were compared; bootstrap resampling grouped all four dialect variants by base question. These intervals are exploratory and conditional on the annotation process. They do not correct possible systematic label error.

4 Results

4.1 Intervention and dataset coverage

The heuristic routed 378 of 400 held-out variants (94.5%) to intervention. Only 22 variants avoided the additional pathway. Thus, selective retrieval was almost always active in the tested configuration. No duplicate held-out rows or logged execution errors were detected; nevertheless, eight answers were blank. The intervention frequency should not be interpreted as a baseline factual-error rate.

4.2 Provisional held-out correctness

Table 2: Provisional condition results (n = 400 each).

Condition	Correct	Ambig.	Eval. acc.	All-set bounds
Baseline	132	74	40.5%	33.0–51.5%
Normalization	122	74	37.4%	30.5–49.0%
Always-RAG	145	65	43.3%	36.3–52.5%
DialectGuard	145	65	43.3%	36.3–52.5%

Table 2 reports the latest AI-assisted and rule-revised labels. The denominator for evaluated accuracy varied with ambiguity, so the columns do not constitute a definitive full-benchmark ranking. The wide full-sample identification ranges overlapped substantially. In particular, the matching 43.3% adjudicable accuracy values for DialectGuard and always-RAG did not establish equivalence over all test inputs.

4.3 Paired comparisons

Table 3: Complete-case paired correctness contrasts.

Contrast	Pairs	Δ (pp)	95% cluster CI
DG − Baseline	294	+3.7	−7.6 to +14.8
DG − Always-RAG	318	0.0	−5.8 to +6.0
RAG − Baseline	292	+2.4	−9.3 to +14.2
Norm. − Baseline	314	−1.9	−7.1 to +3.7

In the 318 adjudicable DialectGuard versus always-RAG pairs, there were 24 improvements, 24 deteriorations, and 270 ties, giving an observed mean correctness difference of zero. The question-cluster bootstrap interval spanned both moderate improvement and deterioration. This analysis provides no reliable evidence of factual superiority and cannot establish statistical equivalence. All paired results were conditional on the subset with determinate labels.

4.4 Risk detector diagnostic

Among 326 held-out variants with determinate baseline correctness labels, the risk detector recorded 193 true positives, 117 false positives, one false negative, and 15 true negatives when “positive” meant a baseline answer judged incorrect. Conditional precision was 62.3%, recall was 99.5%, and specificity was 11.4%. Seventy-four baseline variants had ambiguous correctness labels and were excluded. The very low specificity was consistent with the 94.5% overall high-risk routing rate.

4.5 Dialect sensitivity

Correctness and missing-label coverage varied descriptively by Arabic variety, but no systematic dialect ranking was justified by the small number

of base questions, incomplete annotation, and within-question dependence. Four versions of the same base question are not independent test items. Full condition-by-variety summaries are provided as supplemental research tables in the analysis release and should not be interpreted as prevalence estimates for Arabic-speaking populations.

4.6 Latency and computation

Table 4: Median recorded per-variant latency.

Condition	Median (seconds)
Baseline	1.954
Normalization	4.420
Always-RAG	7.705
DialectGuard	16.491

DialectGuard was slower than always-RAG in 399 of 400 paired held-out variants. The median paired latency difference was approximately 8.735 seconds. A question-cluster analysis of base-level timing differences yielded a median near 8.382 seconds with a bootstrap 95% interval of roughly 8.201–8.572 seconds under the recorded local runtime. These measurements establish an unfavorable latency trade-off for this implementation, not a universal computational lower bound for risk-based retrieval.

5 Discussion

The experiment demonstrated a central difficulty in selective retrieval: the value of routing depends jointly on detecting genuine errors, rejecting correct answers, and computing the risk score cheaply. DialectGuard obtained high conditional recall, but did so by assigning almost every question to the intervention pathway. Its routing stage required outputs from multiple language models. Consequently, a high intervention rate combined with multi-model overhead produced a larger measured latency than simply retrieving on every question. This finding contradicts the initial computational-efficiency hypothesis under the tested local implementation.

The factuality hypothesis was not verified either. On the observed determinate paired subset, DialectGuard did not outperform always-on RAG, and the uncertainty interval was wide. Moreover, the main conditions varied by generator as well as retrieval, while the evidence corpus was incomplete and not independently fact-checked. Thus, assigning any outcome solely to retrieved evidence would overstate the experimental control. The controlled-Qwen experiment and ablations help characterize implementation behavior, but they do not eliminate the confound in the primary held-out comparison.

A further methodological lesson concerns evaluation itself. Arabic synonyms and transliterations make exact-string scoring inappropriate as a sole metric. AI-assisted semantic triage was useful for prioritizing responses, but unresolved ambiguity and unvalidated rule-based reassignment limit inferential strength. Correctness labels cannot directly establish whether each generated claim was supported by retrieved evidence. It would be misleading to convert the present outcomes into a definitive hallucination reduction percentage.

The study has a coherent negative result: response disagreement may identify many likely errors, but a computationally expensive disagreement detector can undermine adaptive retrieval. This result motivates smaller uncertainty predictors, cached inference, single-generator designs, confidence calibration, and cheaper pre-generation routing. These are prospective improvements, not benefits demonstrated by the current system.

6 Limitations and Ethical Considerations

The held-out set comprised 100 source-derived base questions, not a representative sample of all Arabic factual knowledge. Dialectal adaptations were not all independently speaker-certified, and complete semantic-overlap verification was not documented. The evidence collection included 37 relevance-screened passages from Arabic Wikipedia, without independent claim-level factual checks and without guaranteed held-out coverage. Baseline and RAG conditions were confounded by generator identity. Risk measurements were heuristic rather than calibrated probabilities. The fixed local hardware and sequential loading design limit external validity of latency results.

AI-assisted classification and rule-based revisions introduced potential systematic annotation bias. Seventy-four baseline variants and 278 total responses remained ambiguous in the latest proposed dataset; independently verified evidence-support labels were unavailable. Reported significance intervals are exploratory conditional analyses, not estimates corrected for annotation error. These limitations were retained explicitly to avoid overclaiming reliable Arabic fact checking. The repository should not be deployed as an authoritative verification system or used for high-stakes decisions without additional validation.

7 Conclusion

DialectGuard evaluated response-risk-triggered retrieval for Arabic factual question answering

using four linguistic varieties and four primary conditions. The full implementation and experimental record were reproducibly organized, but the proposed method did not realize the hoped-for computational advantage: 94.5% of held-out prompts triggered intervention and measured median latency exceeded always-on RAG. Available AI-assisted factuality labels provided no observed paired correctness advantage over always-on RAG, although unresolved labels and incomplete evidence verification prevented a definitive factuality conclusion. Future research should compare inexpensive calibrated routers using the same generator across all conditions, independently audited evidence, and expert-reviewed dialectal datasets. The present contribution is an empirical account of both the promise and the costs of response-driven adaptive mitigation.

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Appendix A: Configuration and Provenance

The experiment repository is available at https://github.com/MazenZeybek/DialectGuard. The frozen held-out inference revision was 12cd956c5a1afc11b3a156c1e64f8d863e97a09a. The held-out results table, run manifest, quality-control report, model revision file, and experimental configuration form the primary provenance record. The official development evidence collection had 37 relevance-screened passages from 31 source URLs. Historical frozen configuration contents should not be overwritten during repository cleanup.

Appendix B: Annotation Decision Protocol

Correct: the factual answer addresses the question accurately, including acceptable Arabic aliases. Incorrect: the answer contradicts the target fact or provides a wrong required entity. Abstention: no substantive answer is given. Empty: no final answer text is produced. Ambiguous: sufficient information is unavailable to make a defensible decision, including underspecified questions or conflicting reference interpretations. The protocol separates these answer-level classes from claim-level factual support, which must be checked against appropriate independent evidence. Historical dates should be judged using the precise question wording (for example, award year versus ceremony year).

Appendix C: Supplementary Diagnostics

Across the four held-out conditions, the recorded run produced 400 results per condition and eight empty generated final responses. Latest proposed per-condition counts were: baseline 132 correct, 194 incorrect, 74 ambiguous; normalization 122 correct, 204 incorrect, 74 ambiguous; always-RAG 145 correct, 138 incorrect, 47 abstentions, five empty, 65 ambiguous; DialectGuard 145 correct, 127 incorrect, 60 abstentions, three empty, 65 ambiguous. These are proposed annotations, not independent expert ground truth. Further controlled

comparisons, ablations, and supporting raw tables are in the research results package.

Appendix D: Reproduction and Release Caveats

A reproduction should pin the model versions and configurations, preserve the held-out separation, use the recorded evidence corpus, and record hardware and cache behavior. The original pipeline executed on an Apple Silicon M2 system with 16 GB unified memory. Per-question timing figures should not be interpreted as including cold model and index initialization. Public research releases should exclude private gold keys and unlicensed data and should cite the frozen experiment revision. A future publication venue may require an anonymized author version and use of its official style files.